

**M A S A R Y K  
U N I V E R S I T Y**

FACULTY OF INFORMATICS

# **Implementation of MCP Server for Red Hat OpenShift AI Workbenches**

Bachelor's Thesis

**ADAM DAVID MALÝ**

Brno, Fall 2025

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Advisor: Honza + Martin

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MUNI  
FI

## **Declaration**

Hereby I declare that this paper is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

Adam David Malý

**Advisor:** Honza + Martin

## **Abstract**

This thesis presents MCP server implementation for RHOAI workbenches.

## **Keywords**

MCP, RHOAI, Workbench, Server, Implementation

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# Introduction

The Model Context Protocol (MCP) is an open standard that enables AI assistants to interact with external tools and data sources in a structured way. Red Hat OpenShift AI (RHOAI) is a platform for developing, training, and deploying machine learning models on OpenShift clusters. Within RHOAI, *workbenches* provide interactive, notebook-based development environments for data scientists.

This thesis presents the design and implementation of an MCP server that exposes RHOAI workbench management capabilities to MCP-compatible AI clients. The server allows AI assistants to create, list, start, stop, and delete workbenches through a standardised protocol interface, enabling conversational and automated management of data science environments.

The thesis is structured as follows. Chapter ?? introduces the necessary background on the MCP protocol and the RHOAI platform. Chapter ?? presents the requirements analysis and architecture design of the MCP server. Chapter ?? describes the implementation in detail. Chapter ?? covers testing and evaluation of the result. The thesis concludes with a summary and directions for future work.

# 1 Background

## 1.1 Model Context Protocol

The Model Context Protocol (MCP) is an open protocol introduced by Anthropic in 2024. It defines a standard interface through which AI language model clients can discover and invoke tools, read resources, and receive structured responses from external servers. MCP uses a JSON-RPC 2.0 transport layer and supports both local (stdio) and remote (HTTP/SSE) communication modes.

An MCP server exposes a set of *tools* – callable functions with defined input schemas – that a connected AI client can invoke. This makes it possible to extend the capabilities of AI assistants with domain-specific actions without modifying the model itself.

## 1.2 Red Hat OpenShift AI

Red Hat OpenShift AI (RHOAI) is a machine learning platform built on top of OpenShift, Red Hat's enterprise Kubernetes distribution. It provides a managed environment for data scientists and ML engineers, including model serving, pipelines, and interactive development environments.

### 1.3 Workbenches

A workbench in RHOAI is a containerised Jupyter notebook environment provisioned as a Kubernetes Notebook custom resource. Each workbench is associated with a notebook image, a hardware profile specifying compute resources, and one or more persistent volume claims for storage. Workbenches can be started, stopped, and deleted through the RHOAI dashboard or directly via the Kubernetes API.

```
![\logo](fithesis/logo/mu/fithesis-fi-color "The old-style seal of the  
Faculty of Informatics")
```

|       |      |         |        |    |    |    |    |     |     |     |     |   |   |   |   |
|-------|------|---------|--------|----|----|----|----|-----|-----|-----|-----|---|---|---|---|
| Right | Left | Default | Center | 12 | 12 | 12 | 12 | 123 | 123 | 123 | 123 | 1 | 1 | 1 | 1 |
|-------|------|---------|--------|----|----|----|----|-----|-----|-----|-----|---|---|---|---|

: This is a table with different types of alignment.

This is a bullet list. Unlike numbered lists, bulleted lists contain an **unordered** set of bullet points. When a bullet point contains multiple paragraphs, the list is typeset as follows:

- \* The first item of a bullet list  
that spans several paragraphs, \* the second item of a bullet list, \* the third item of a bullet list.

When none of the bullet points contains multiple paragraphs, the list has a more compact form:

- \* The first item of a bullet list, \* the second item of a bullet list, \* the third item of a bullet list.

Unlike a bulleted list, a numbered list implies chronology or ordering of the bullet points. When a bullet point contains multiple paragraphs, the list is typeset as follows:

1. The first item of an ordered list  
that spans several paragraphs, 2. the second item of an ordered list, 3. the third item of an ordered list. . If you are feeling lazy, . you can use hash enumerators as well.

When none of the bullet points contains multiple paragraphs, the list has a more compact form:

6. The first item of an ordered list, 7. the second item of an ordered list, 8. the third item of an ordered list.

Definition lists are used to provide definitions of terms. When a definition contains multiple paragraphs, the list is typeset as follows:

Term 1

: Definition 1

\*Term 2\*

: Definition 2

Some code, part of Definition 2

Third paragraph of Definition 2.

When none of the bullet points contains multiple paragraphs, the list has a more compact form:

Term 1 : Definition 1 \*Term 2\* : Definition 2

Block quotations are used to include an excerpt from an external document in way that visually clearly separates the excerpt from the rest of the work:

> This is the first level of quoting. > > > This is nested blockquote.  
> > Back to the first level.

Footnotes are used to include additional information to the document that are not necessary for the understanding of the main text. Here is a footnote reference<sup>[Here is the footnote.] and another.<sup>[longnote]</sup>  
<sup>[longnote]</sup> : Here's one with multiple blocks.</sup>

Subsequent paragraphs are indented to show that they belong to the previous footnote.

Some code

The whole paragraph can be indented, or just the first line. In this way, multi-paragraph footnotes work like multi-paragraph list items.

Citations are used to provide bibliographical references to other documents. This is a regular citation [ @borgman03, p. 123 ]. This is an in-text citation: @borgman03. You can also cite several authors at once using both regular [ see @borgman03, p. 123; @greenberg98, sec. 3.2; and @thanh01 ] and in-text citations: @borgman03 [ p.123; @greenberg98, sec. 3.2; @thanh01 ].

Code blocks are used to include source code listings into the document:

```
include <stdio.h> include <unistd.h> include <sys/types.h> include <sys/wait.h> // This is a comment
int main(int argc, char **argv) while (-c > 1 !fork()); sleep(c = atoi(v[c])); printf("wait(0);
return 0;
```

There is an alternative syntax for code blocks that allows you to specify additional information, such as the language of the source code. This information can be used for syntax highlighting:

```
''' sh !/bin/sh fac() if [ "1" -leq 1 ]; then echo 1 else echo ("1" * fac(("1" -
1)))) fi'''
```

```
ruby Here's a way to empty an array. joe = [ 'eggs.', 'some',
'break', 'to', 'Have' ] print(joe.pop, " ") while joe.size > 0 print ""
```

## 2 These are

### 2.1 the available

#### 2.1.1 sectioning

commands.

Paragraphs and

**subparagraphs are available as well.** Inside the text, you can also use unnumbered lists,

- such as
- this one
  - and they can be nested as well.
  - » You can even turn the bullets into something fancier,
  - § if you so desire.

Numbered lists are

1. very
  - (a) similar

and so are description lists:

**Description list** A list of terms with a description of each term

The spacing of these lists is geared towards paragraphs of text. For lists of words and phrases, the paralist package offers commands

- that
  - are
  - \* better
  - suited
- 1. to
  - (a) this
    - i. kind of

A. content.

The `amsthm` package provides the commands necessary for the type-setting of mathematical definitions, theorems, lemmas and proofs.

**Theorem 2.1.1.** *This is a theorem that offers a profound insight into the mathematical sectioning commands.*

**Theorem 2.1.2** (Another theorem). *This is another theorem. Unlike the first one, this theorem has been endowed with a name.*

**Lemma 2.1.3.** *Let us suppose that  $x^2 + y^2 = z^2$ . Then*

$$\left\langle u \left| \sum_{i=1}^n F(e_i, v) e_i \right. \right\rangle = F\left(\sum_{i=1}^n \langle e_i | u \rangle e_i, v\right). \quad (2.1)$$

*Proof.*  $\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$ . □

**Corollary 2.1.4.** *This is a corollary.*

*Remark.* This is a remark.

### 3 Floats and references

The logo of the Masaryk University is shown in Figure 5.1 and Figure 5.2 at pages 10 and 11. The weather forecast is shown in Table 5.1 at page 11. The following chapter is Chapter 6 and starts at page 12. Items 3, 3b, and 3(c)iv are starred in the following list:

1. some text
2. some other text
3. ★
  - (a) some text
  - (b) ★
  - (c) some other text
    - i. some text
    - ii. some other text
    - iii. yet another piece of text
    - iv. ★
  - (d) yet another piece of text
4. yet another piece of text

If your reference points to a place that has not yet been typeset, the `\ref` command will expand to `??` during the first run of `pdflatex fi-pdflatex.tex` and a second run is going to be needed for the references to resolve. With online services – such as Overleaf – this is performed automatically.

The logo of Masaryk University, consisting of the words "MASARYK" and "UNIVERSITY" stacked vertically in a bold, blue, sans-serif font.

**Figure 3.1:** The logo of the Masaryk University at 6.3 cm



**Figure 3.2:** The logo of the Masaryk University at  $\frac{1}{2}$  and  $\frac{1}{3}$  of text width

**Table 3.1:** A weather forecast

| Day       | Min Temp | Max Temp | Summary                                                      |
|-----------|----------|----------|--------------------------------------------------------------|
| Monday    | 13°C     | 21°C     | A clear day with low wind and no adverse current advisories. |
| Tuesday   | 11°C     | 17°C     | A trough of low pressure will come from the northwest.       |
| Wednesday | 10°C     | 21°C     | Rain will spread to all parts during the morning.            |

## 4 Mathematical equations

T<sub>E</sub>X comes pre-packed with the ability to typeset inline equations, such as  $e^{ix} = \cos x + i \sin x$ , and display equations, such as

$$\mathbf{A}^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

L<sup>A</sup>T<sub>E</sub>X defines the automatically numbered equation environment:

$$\gamma Px = PAx = PAP^{-1}Px. \quad (4.1)$$

The package `amsmath` provides several additional environments that can be used to typeset complex equations:

1. An equation can be spread over multiple lines using the `multline` environment:

$$\begin{aligned} a + b + c + d + e + f + b + c + d + e + f + b + c + d + e + f \\ + f + g + h + i + j + k + l + m + n + o + p + q \end{aligned} \quad (4.2)$$

2. Several aligned equations can be typeset using the `align` environment:

$$a + b = c + d \quad (4.3)$$

$$u = v + w + x \quad (4.4)$$

$$i + j + k + l = m \quad (4.5)$$

3. The `alignat` environment is similar to `align`, but it doesn't insert horizontal spaces between the individual columns:

$$a + b + c + d \quad = 0 \quad (4.6)$$

$$e + f + g = 5 \quad (4.7)$$

4. Much like chapter, sections, tables, figures, or list items, equations – such as (6.8) and (My equation) – can also be labeled and referenced:

$$b_{11}x_1 + b_{12}x_2 + b_{13}x_3 \quad = y_1, \quad (4.8)$$

$$b_{21}x_1 + b_{22}x_2 \quad + b_{24}x_4 = y_2. \quad (\text{My equation})$$

5. The `gather` environment makes it possible to typeset several equations without any alignment:

$$\psi = \psi\psi, \tag{4.9}$$

$$\eta = \eta\eta\eta\eta\eta\eta, \tag{4.10}$$

$$\theta = \theta. \tag{4.11}$$

6. Several cases can be typeset using the `cases` environment:

$$|y| = \begin{cases} y & \text{if } z \geq 0, \\ -y & \text{otherwise.} \end{cases} \tag{4.12}$$

For the complete list of environments and commands, consult the `amsmath` package manual<sup>1</sup>.

---

1. See <http://mirrors.ctan.org/macros/latex/required/amsmath/amslldoc.pdf>. The `\url` command is provided by the package `url`.

## 5 We have several FONTS *at disposal*

The serifed roman font is used for the main body of the text. *Italics are typically used to denote emphasis or quotations.* The teletype font is typically used for source code listings. The **bold**, SMALL-CAPS and sans-serif variants of the base roman font can be used to denote specific types of information.

We can also change the font size, although it is usually not necessary.

A wide variety of mathematical fonts is also available, such as:

$ABC, \mathcal{ABC}, \mathbf{ABC}, \text{ABC}, \textit{ABC}, \text{ABC}$

By loading the amsfonts packages, several additional fonts will become available:

$\mathfrak{ABC}, \mathbb{ABC}$

Many other mathematical fonts are available<sup>1</sup>.

---

1. See <http://tex.stackexchange.com/a/58124/70941>.

## 6 Inserting the bibliography

After linking a bibliography database files to the document using the `\thesissetup{bib={file1,file2,...}}` command, you can start citing the entries. This is just dummy text borgman03 lightly sprinkled with citations [p. 123]greenberg98. Several sources can be cited at once: [?, ?, ?]. greenberg98 was written by greenberg98 in greenberg98. We can also produce greenberg98 or (greenberg98, greenberg98). The full bibliographic citation is: *greenberg98*. We can easily insert a bibliographic citation into the footnote greenberg98.

The `\nocite` command will not generate any output, but it will insert its arguments into the bibliography. The `\nocite{*}` command will insert all the records in the bibliography database file into the bibliography. Try uncommenting the command and watch the bibliography section come apart at the seams.

When typesetting the document for the first time, citing a work will expand to [work] and the `\printbibliography` command will produce no output. It is now necessary to generate the bibliography by running `biber fi-pdflatex.bcf` from the command line and then by typesetting the document again twice. During the first run, the bibliography section and the citations will be typeset, and in the second run, the bibliography section will appear in the table of contents.

The `biber` command needs to be executed from within the directory, where the  $\text{\LaTeX}$  source file is located. In Windows, the command line can be opened in a directory by holding down the Shift key and by clicking the right mouse button while hovering the cursor over a directory. Select the Open Command Window Here option in the context menu that opens shortly afterwards.

With online services – such as Overleaf – or when using an automatic tool – such as  $\text{\LaTeX}$ MK – all commands are executed automatically. When you omit the `\printbibliography` command, its location will be decided by the template.

[heading=bibintoc]

## 7 Inserting the index

After using the `\makeindex` macro and loading the `makeidx` package that provides additional indexing commands, index entries can be created by issuing the `\index` command. It is possible to create ranged index entries, which will encompass a span of text. To insert complex typographic material – such as  $\alpha$  or  $\TeX$  – into the index, you need to specify a text string, which will determine how the entry will be sorted. It is also possible to create hierarchical entries.

After typesetting the document, it is necessary to generate the index by running

```
texindy -I latex -C utf8 -L <locale> fi-pdflatex.idx
```

from the command line, where *<locale>* corresponds to the main locale of your thesis – such as `english`, and then typesetting the document again.

The `texindy` command needs to be executed from within the directory, where the  $\LaTeX$  source file is located. In Windows, the command line can be opened in a directory by holding down the Shift key and by clicking the right mouse button while hovering the cursor over a directory. Select the Open Command Window Here option in the context menu that opens shortly afterwards.

With online services – such as Overleaf – the commands are executed automatically, although the locale may be erroneously detected, or the `makeindex` tool (which is only able to sort entries that contain digits and letters of the English alphabet) may be used instead of `texindy`. In either case, the index will be ill-sorted.

## **A An appendix**

Here you can insert the appendices of your thesis.